94-844 Generative AI Lab

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Technology Instructions for ASSIGNMENT 2 – Retrieval Augmented Generation (RAG) with LLMs.

Please make sure to upgrade your Python version to at least 3.12 to complete this assignment.

- **Instructions for updating Python:** https://docs.google.com/document/d/1kONEZ562H97ZLpYNZrn2KcxDVI9yUrh5LEX09DYCkvk/edit?usp=sharing

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Required Canvas Downloads:

Please download these materials to complete the Homework 2 assignment:

- HW02_Assignment.pdf
- HW02_Homework_Questions.pdf
- hw02-technology-instructions.pdf
- HW02-notebook.ipynb
- CMU-student-policy-handbook.pdf
- Policy-Question-Dataset.CSV
- requirements.txt

FYI on Locally Hosting LLMs and LLM APIs:

You are welcome to locally host or use any LLM API that is permissible to use in the United States for this homework assignment.

Required Accounts and API Keys:

Strongly recommended: Please consider creating a **pseudonym email address** to use with the HuggingFace or other platforms you might use for homework assignments. We ask that you consider doing this to protect your federally protected student data.[1]

Please create accounts and API keys for the following, if you don't already have them:

- **HuggingFace:**
  - Sign-up for HuggingFace account: https://huggingface.co/join
  - How to get API tokens: https://huggingface.co/docs/hub/en/security-tokens

Choosing and Setting Up Your Computing Environment:

Below, we provide instructions for working with the HW2 Python/Jupyter Notebook in VS Code and a HuggingFace Space. **Please note that if you decide to work in a HuggingFace Space, it is not free.**

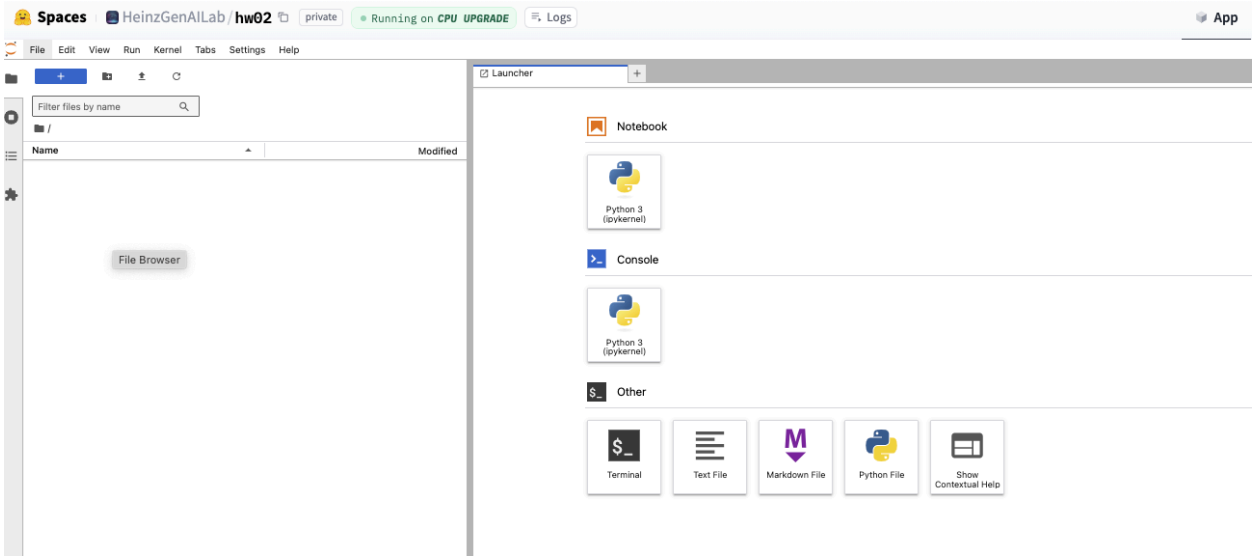
OPTIONAL → Working in a HuggingFace Space:

Please do NOT use your ANDREW ID or EMAIL or PERSONAL NAME if you create a HuggingFace Account for this assignment. We ask that you do NOT do this for student data privacy reasons. Instead, please create a pseudonym for yourself and use that if needed. Please note that doing your homework assignment on HuggingFace is not recommended because we cannot guarantee that it complies with U.S. student data privacy rules. However, if you prefer to use it with this risk in mind, here are the instructions:

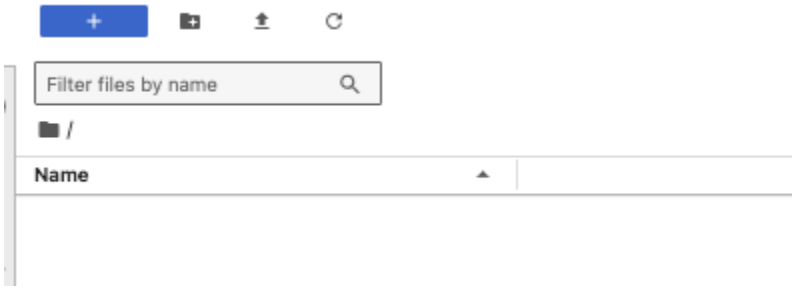
1. Navigate to HuggingFace Spaces at https://huggingface.co/spaces

[1] For more information, please look up student data protection rights under Family Educational Rights and Privacy Act (FERPA].

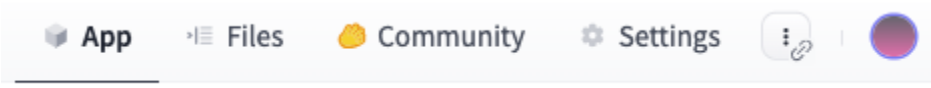
2. Click on the “Create new Space” button.
3. **Fill out the form**:
    a. Space name: the name you want to give your space. For example, “homework 2”
    b. Short description: “computing space for homework 2” or similar.
    c. License: any you choose.
4. **Select the Space SDK**:
    a. Please click the Docker button.
    b. Then click the JupyterLab button as the Docker Template.
5. **Space Hardware:**
    a. Please change to “CPU upgrade 8 vCPU 32 GB $0.03 per hour”
6. **Persistent storage**: we recommend using the default storage.
7. Space Secrets: you can add a password to your space by writing a password in the box that says “add secret” on the line that says JUPYTER TOKEN.
8. **Privacy**: we recommend making the space private but you can make it public if you like.
9. Click “create space button”
10. You will be navigated to a screen where you will see the logs for the space build. Wait on this screen until the space is done building.
11. Once the space is built, log into it. You should see:



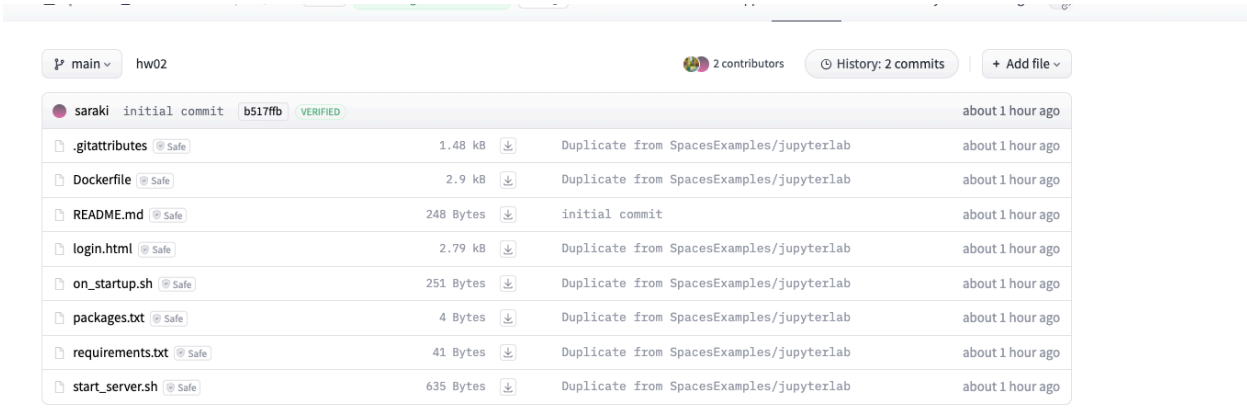
12. Click the up-arrow button on the top of the left-hand side bar, depicted below.



13. Upload these materials, which you will need to have downloaded from Canvas to your computer:
    a. HW02_Assignment.pdf
    b. HW02_Homework_Questions.pdf
    c. hw02-technology-instructions.pdf
    d. HW02-notebook.ipynb
    e. CMU-student-policy-handbook.pdf
    f. Policy-Question-Dataset.CSV
    g. requirements.txt
14. Now click on the "Files" button at the top right-hand side of the interface:



    a. You should see a file list that looks like this:



    b. Click on the file named "requirements.txt"
    c. Click the "edit" button.
    d. Copy and paste the requirements listed in the "requirements.txt" that you downloaded from Canvas into the Spaces requirements.txt.
    e. Press the "commit changes to main" button.
15. Navigate back to the Spaces interface.
16. Now click on the "Settings" button at the top right-hand side of the interface:
    a. Scroll down to the section named "Restart this Space" and press the "Restart space" button.
    b. This will install the Python requirements you need to complete Homework 2 to the space. By doing this, you should not need to install the requirements again. Or, the requirements will be installed each time you restart your space or Space kernel.

17. Navigate back to your Space's main interface.
18. Click on the "HW02-notebook.ipynb" file.
19. A jupyter notebook should open.
20. Begin computing.
21. Email Teaching team (Professors, TAs) with any questions or if you experience challenges setting up your environment.

Working in VS Code:

Homework 2 was designed for working in VS Code with the Python/Jupyter Notebook. Please email Sara Kingsley <skingsle@andrew.cmu.edu> if you have any questions or do not have access to VS Code.

Important Note re. the Virtual Environment for this assignment:

For this assignment, you will need to create a new virtual environment, e.g. please do not use the virtual environment you created for lab 1 or homework assignment 1. *The Homework / Lab 1 virtual environment uses a different version of Python and does not have the required Python dependencies required for Homework 2 installed.* Instructions for creating a new virtual environment for this Homework 2 assignment are provided below.

Important Note for Windows Users:

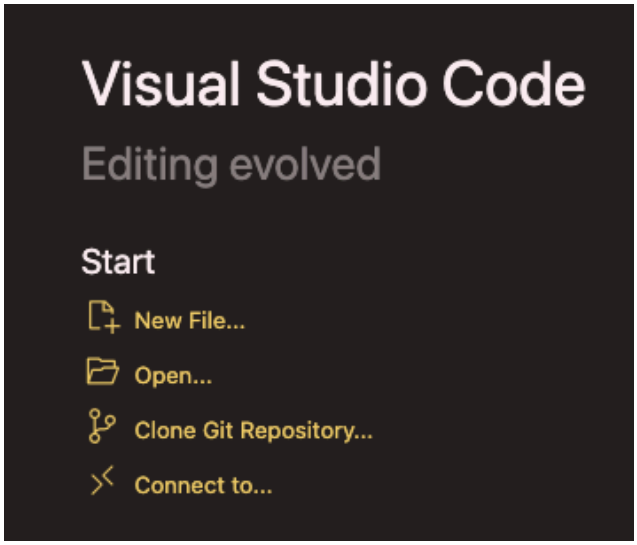
Note, for this assignment, if you are a Windows user, you might need to install Ubuntu first in order to be able to use a Bash shell in VS Code. This requires installing "wsl". Directions to install WSL are here:
https://www.pcworld.com/article/794691/how-to-install-bash-on-windows-11.html

Instructions for Creating a Virtual Environment and installing the Python Requirements:

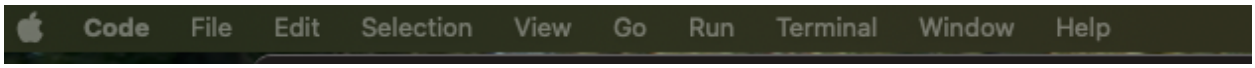
These instructions are specific to VS Code.

1. **If you are using a Python version older than version 3.12, please ensure to update Python to at least Python=3.12**
    a. **Instructions on how to update Python: https://docs.google.com/document/d/1kONEZ562H97ZLpYNZrn2KcxDVI9yUrh5LEX09DYCkvk/edit?usp=sharing**
2. Tutorial materials (not required): *If you would like to practice building a RAG system using the RAG tutorial Python notebook and data, please download the Tutorial materials from Canvas.*
3. **To work on the Homework 2 assignment, please download from Canvas these files:**
    a. HW02_Assignment.pdf

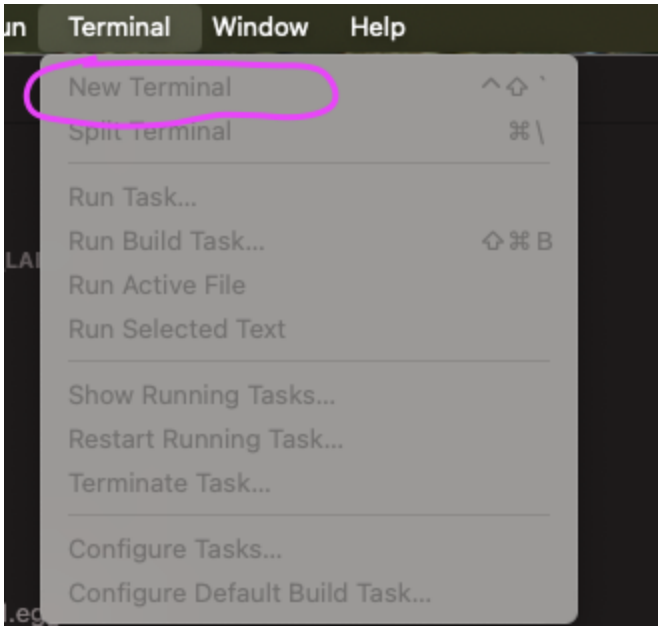
   b. HW02_Homework_Questions.pdf
   c. hw02-technology-instructions.pdf
   d. HW02-notebook.ipynb
   e. CMU-student-policy-handbook.pdf
   f. Policy-Question-Dataset.CSV
   g. requirements.txt
4. Open VS code.
5. A "welcome" tab should appear. Click on the "Open…" option (shown below)



6. On the left-hand side of your VS Code window, look for a symbol that looks like two pieces of paper ( ) and click on it.
7. You should now see a list that gives the names of all the files in your directory (e.g. folder). Keep this view open. I'll ask you to click on a few files, using it.



8. Look to the top of your computer screen, and click on the menu option named "Terminal."
   a. Click on "New Terminal"



9. A Terminal window should pop up at the bottom of your VS Code window.
10. In the Terminal window, type (and then press enter):

```
cd <insert name of your folder/directory>
```

11. Now, look once more at the left-hand side of your VS Code window at the list of files in your directory (e.g. folder).
    a. Check to make sure that the file named "requirements.txt" is there.
    b. If it is not, email Sara.

12. Create a virtual environment:
    a. In your terminal, type:

```
None
python3 -m venv myenv
```

    b. Then type:

MacOS users:

```
None
source myenv/bin/activate
```

Windows users:

```
.\venv\Scripts\activate
```

13. Return to your Terminal window, and type the command below (and then press enter):

```
None
pip install -r requirements.txt
```

This should install the Python packages and their dependencies that you need to have installed on your computer to complete the Homework 2 assignment.

14. Now, look once more at the left-hand side of your VS Code window at the list of files in your directory (e.g. folder).
    a. Check to make sure that the file named "HW02-notebook.ipynb" is there.
    b. If it is not, email Sara.
    c. Click on the file named "HW02-notebook.ipynb". It should open.

15. Run all the custom functions at the start of the notebook (this will define the custom functions you'll need to use in the homework section of the notebook)

16. Navigate to the Homework section of the notebook and complete the coding tasks and homework questions.

Homework Submission Requirements:

1. **(required) LLM Judge Python Code**:
    a. **(required)** You are required to submit the code for your LLM Judge.
    b. (optional) For all other code, such as for your vector database, you are not required to submit your code. You may include other code, but please note the teaching team will not review it for grading purposes.

c. **Please do NOT answer the homework questions in the Python notebook and only submit the Python notebook.** The teaching team will not review your Python notebook to grade your answers.

2. **(required) Homework Report:** You are required to submit a PDF of your written responses to the homework questions.

3. **(required) Spreadsheet:** You are required to copy this Google spreadsheet template and provide copies of the responses to your queries that were submitted to the vector store in it. https://docs.google.com/spreadsheets/d/1T2_fV0LKw9s_BEDKAvYZKKugDD0iBKSUlsnaui3Dmlg/edit?gid=0#gid=0

4. **(required) Presentation Slides:** You are required to create 3-5 presentation slides and submit those to Canvas with your homework submission report PDF.

Accessing the Generative AI models for the Bonus Tasks:
To complete the Bonus tasks / questions, you may need access to Google's Gemma and Meta's Llama models. To get access, you will need to first log into your HuggingFace account. Second, you will need to request access to the models at the links below. Note that it can take up to 24 hours or more for Google or Meta to grant you access to the models. Thus, please request access to the models at least 3 days before the homework is due.

- Request access to Google's Gemma, here: https://huggingface.co/google/gemma-7b
- Request access to Meta's Llama 3.2 1B parameter model, here: https://huggingface.co/meta-llama/Llama-3.2-1B

Note: You may choose to use different models for the Bonus tasks. Please cite the name and link to the models you decide to use from HuggingFace or other API or locally hosting solution. Thank you!